

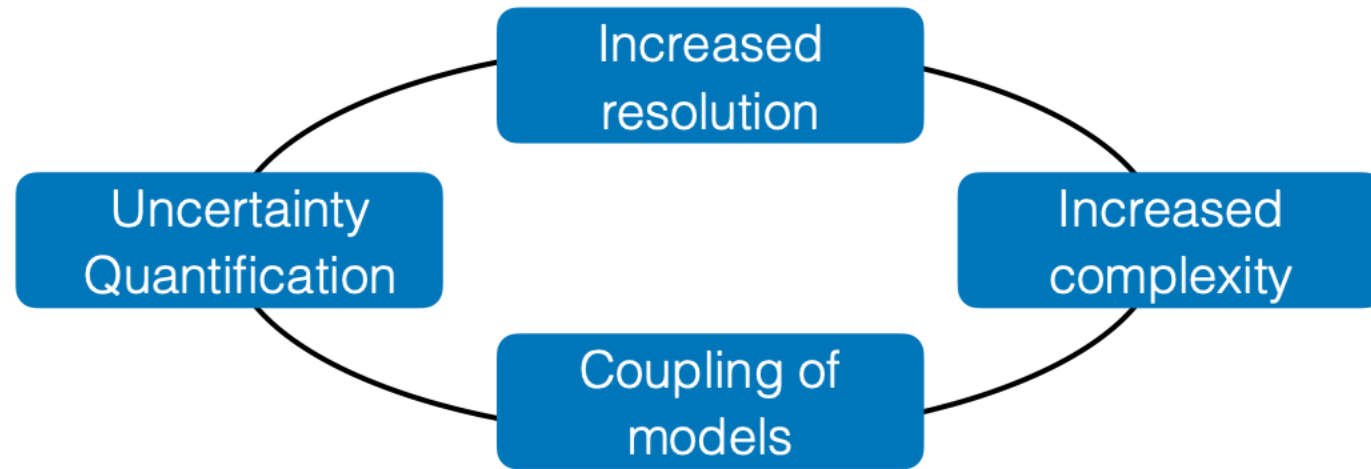
UK HPC and Data facilities

Mark Wilkinson
Director, STFC DiRAC HPC Facility

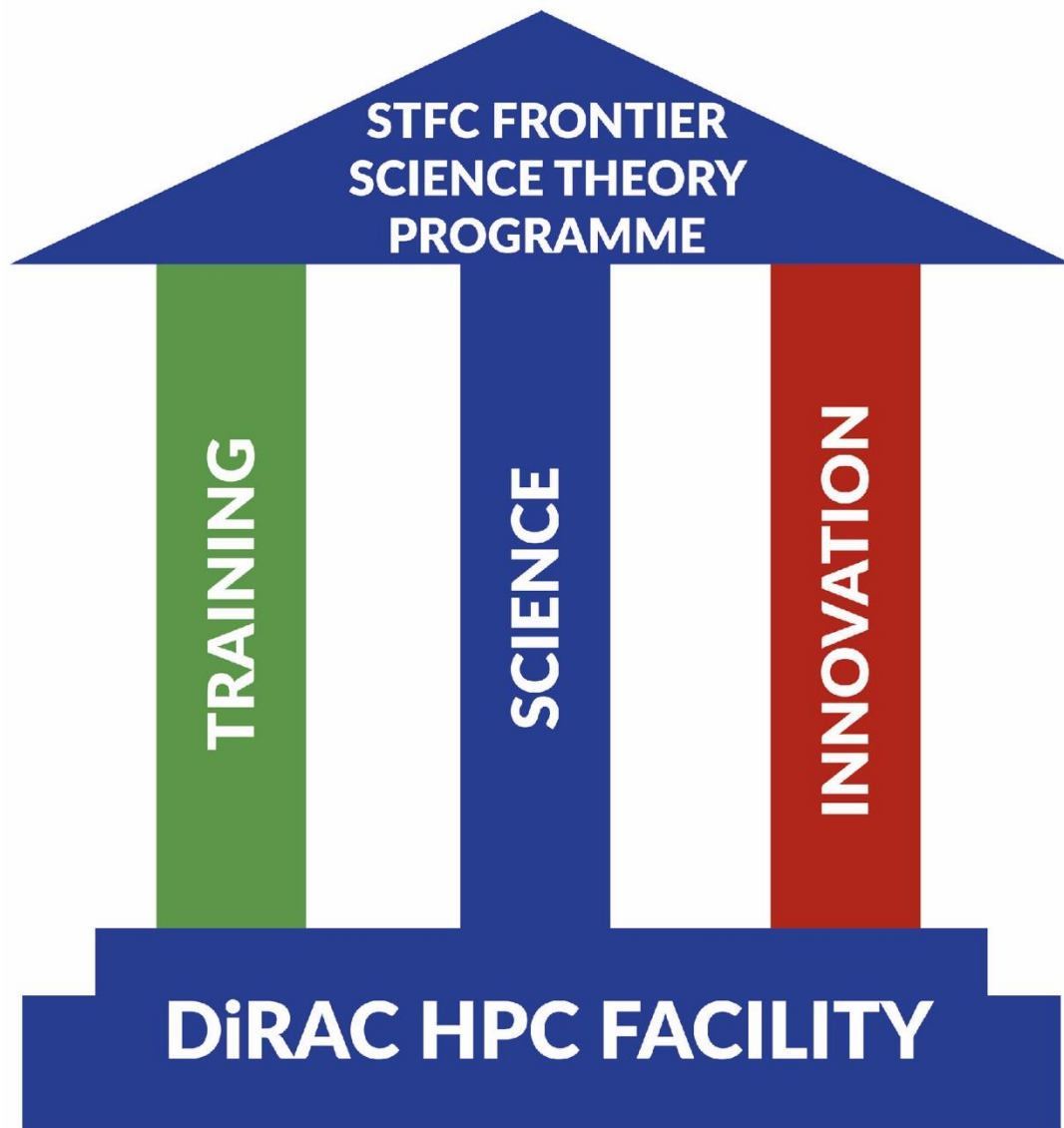
DISCNet High Performance Computing training course
16th May 2025

HPC in research

- Numerical simulation and modelling is core research activity in many fields
 - High Performance Computers are main **scientific instruments** for theory research
- Computational requirements of models increasing due to:



- Constant process of refining and re-defining our tools
 - Convergence of AI and simulations in terms of hardware requirements
- Not just hardware - software and people are key
 - Wide range of skill sets needed, including RSE, DevOps, ResOps, etc.



DiRAC Objectives

- **Scientific Excellence**
 - Support ground-breaking science
 - Enhance UK scientific leadership
- **Investing in people**
 - Attract international researchers
 - Train people
- **UK-wide innovation**
 - Drive UK-wide innovation
 - Prepare for the UKRI Digital Research Infrastructure

- All three strands of DiRAC activity support the STFC Theory programme
- Training and innovation also support broader STFC/UKRI/DSIT objectives

The DiRAC HPC Facility

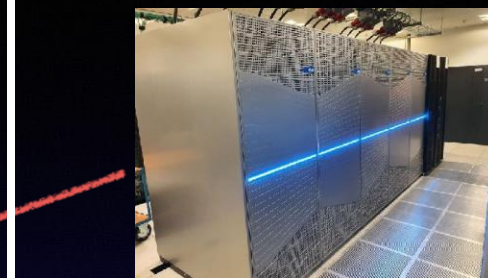
Memory Intensive “COSMA8” (Durham)

- 528 TB RAM
- Large-scale cosmological simulations



Extreme Scaling “Tursa” (Edinburgh)

- 704 Nvidia A100 GPUs
- Large lattice-QCD simulations



Data Intensive “DiaL” (Leicester)

- Heterogeneous architecture for complex simulation and modelling workflows



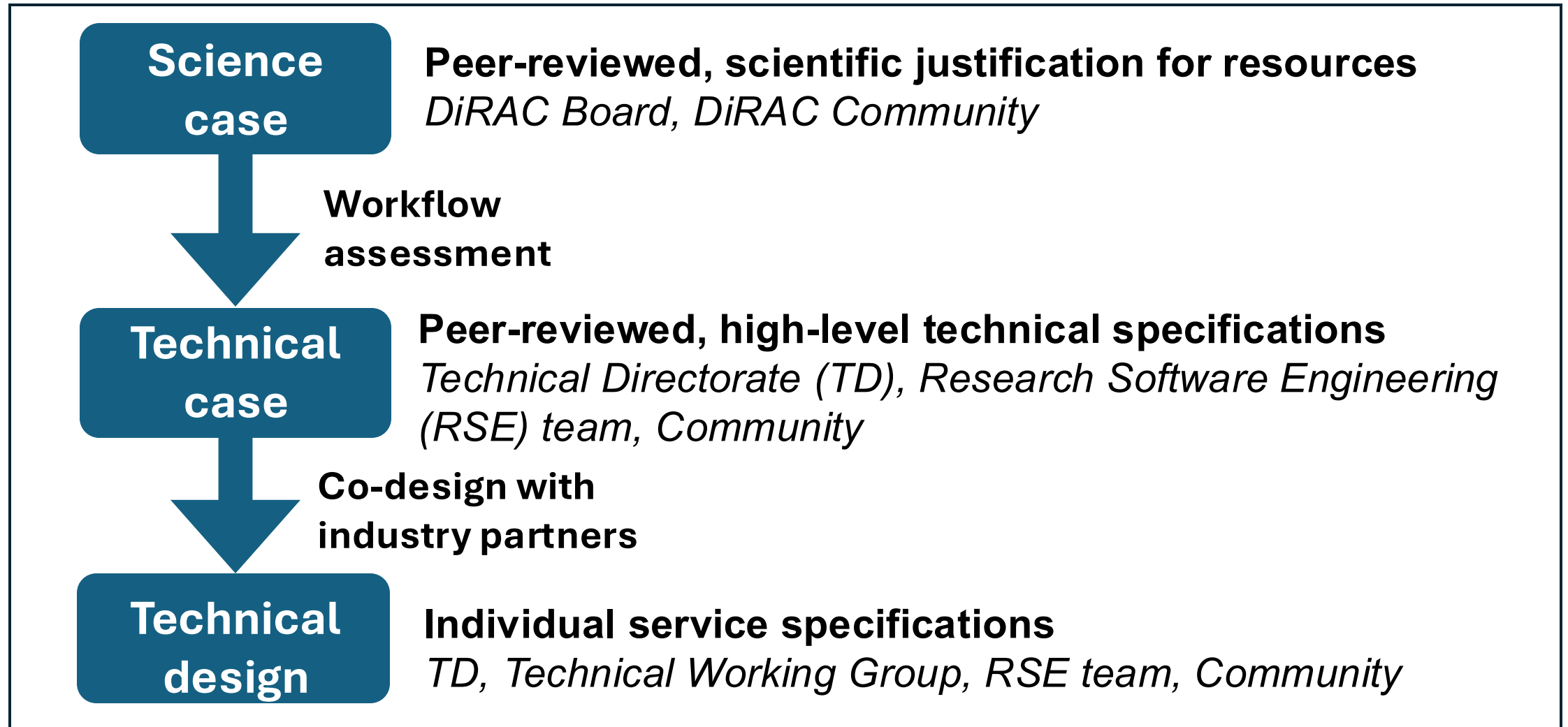
Data Intensive “CSD3” (Cambridge)

- Heterogeneous architecture for complex simulation and modelling workflows



Facility Office (UCL)

Co-design of DiRAC services – a multi-stakeholder partnership

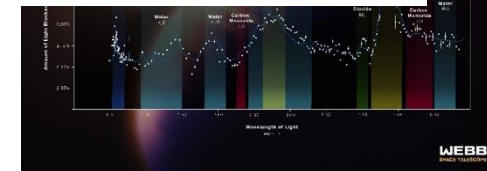
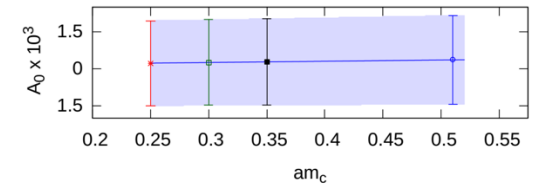
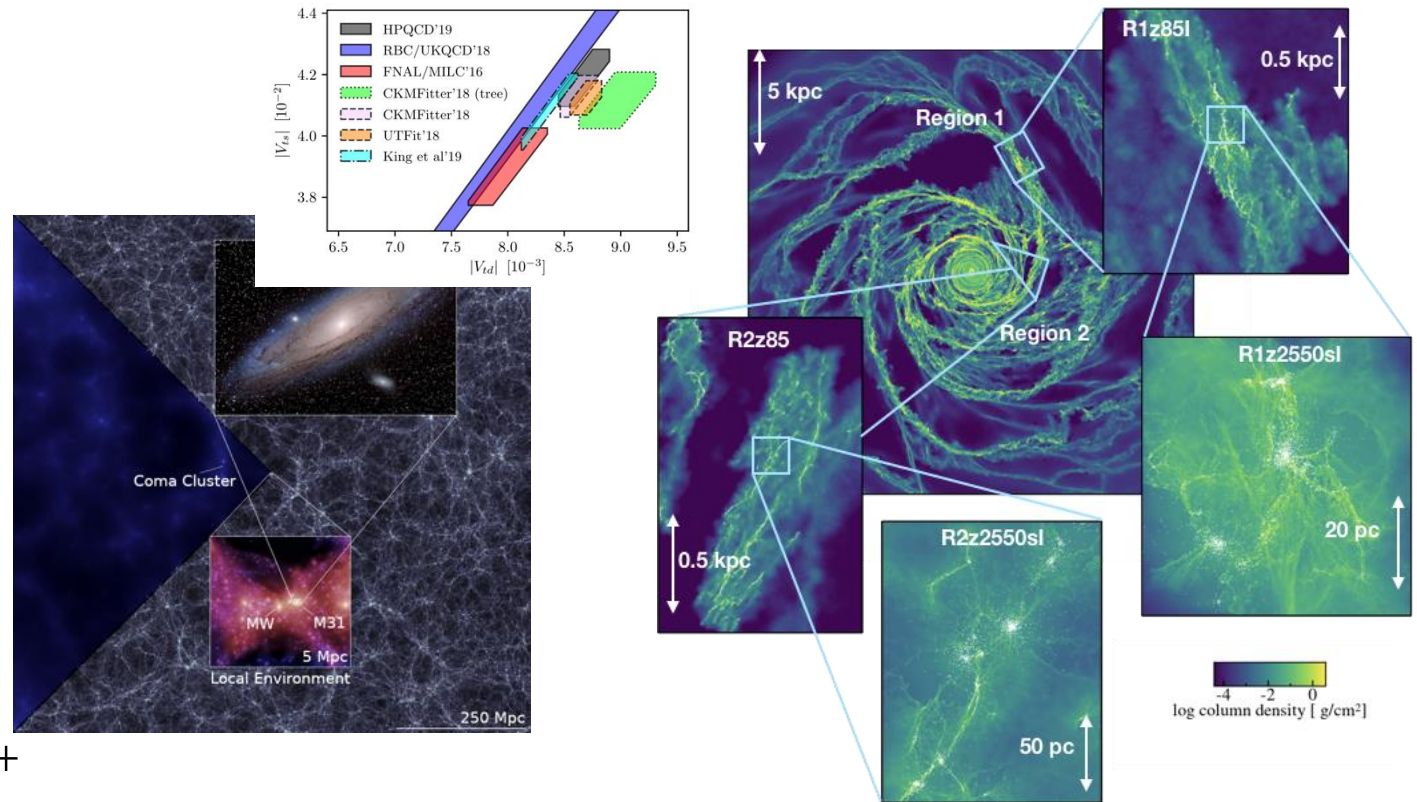


- Science case determines *both* scale and design of DiRAC services
- Design goal to increase arithmetic intensity

DiRAC Science Programme 2024-28

DiRAC

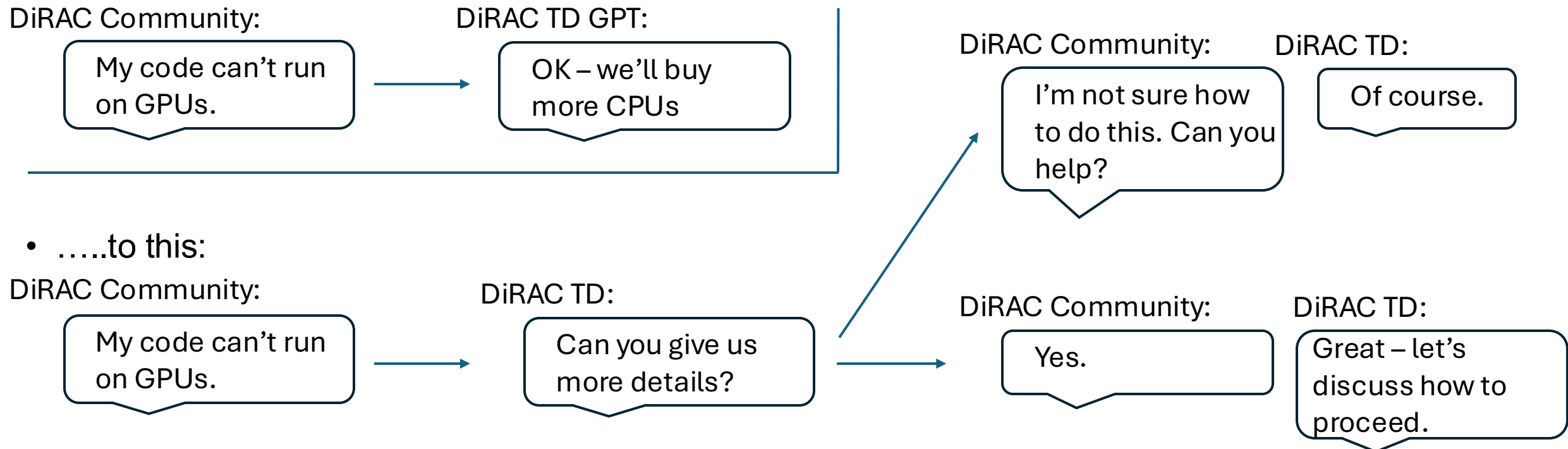
- Capability calculations include:
 - Galaxy formation
 - Lattice Quantum Field Theory
- Data Intensive calculations:
 - Gravitational waves
 - Gaia modelling
 - Precision cosmology
 - Planetary atmospheres
- Data challenges growing rapidly
 - Individual simulations generate 10Pb+
- Increasing use of AI/ML techniques to enhance simulation methods
 - At least 50% of fields are using or exploring AI over next 4 years.
 - DiRAC simulation data can be used to train AI models.
- STFC facilities also use AI extensively for data acquisition, data processing, data analysis



“Trust, but verify”

Evidencing the DiRAC-4 technical case

- Evidence-based development of compute services is at the core of DiRAC’s design philosophy
- The case for DiRAC-4 will be carefully scrutinised by funders (STFC, UKRI DRI, DSIT) and must be as robust as possible
- Assertions made without data to back them up will undermine the whole case
- Need to move conversations from this.....



STFC Exascale Requirements

- Four key attributes of STFC requirements:
 - Heterogeneity and cost-effectiveness
 - Centrality of data
 - Urgency
 - Dependent on investments in people
- Distinctive feature of STFC projects is generation of large-scale data followed by processing, reducing, analysis and/or modelling
 - Data generated by experiments (e.g LHC, DIAMOND, ISIS, etc), survey astronomy facilities (e.g. LSST, SKA), or simulations (e.g. DiRAC).
 - Data volume, quality and velocity are all critical
 - Includes need to move large volumes of data from generation/collection sites to compute centres
- Scale of STFC requirements are significant component of overall UKRI requirement:
 - Current DiRAC resources are 70% the size of Archer2 in terms of flop/s (40% in terms of nodes)
 - GridPP is 12% of WLCG which is already >1EF of compute
 - New UKRI Isambard 3 Tier 2 service is <15% the size of DiRAC CPU provision

Heterogeneous computing

- STFC communities have clearly defined, peer-reviewed and funded science programmes with well-defined computing requirements.
- These requirements are best supported via heterogeneous solutions, both technically and economically:
 - Heterogeneity covers more than just CPU/GPU - e.g. network, data requirements etc.
 - The design of any exascale services should be based on the use cases to be supported rather than a target of 1 EF of LinPack performance
- UKRI exascale planning needs to incorporate this diversity of requirements to avoid severely damaging STFC's science programme.
- This diversity is an asset that is key to the UK's scientific leadership.

Co-design benefits & the importance of people

- Investment in people is vital for productive HPC services
- DiRAC services require specialist technical support for hardware and users
- RSE team supports code improvement and re-factoring, energy efficiency, co-design, procurement & training

Evolution of Grid code (Boyle et al.) performance on Tursa relative to Tesseract

Stage	1 node	% inc.	16 nodes	% inc.	speed up 512 tess
Measured	9.2	-	5.3	-	1.1
Committed	9.2	-	5.83	10%	1.22
Acceptance	9.65	5%	6.15	16%	1.28
Commissioning	12	30%	8.8	66%	1.83
Peak	12.9	40%	9.9	87%	2.06

*James
Richings et al.*

- Tursa Extreme Scaling service (DiRAC@Edinburgh) provides ~9x the performance of its CPU-based predecessor for lattice QCD codes but uses just 0.8x the power.
- Cosma8 Memory Intensive service (DiRAC@Durham) is 4x more efficient for cosmological simulations than comparable systems in Europe – DiRAC services allow us to remain internationally leading

And it's about more than just compute....

Data management is often more challenging

FAIR data principles

All publicly-funded research data should be:

1. Findable
2. Accessible
3. Interoperable
4. Reusable

The EAGLE simulation database

Virgo
Consortium

EAGLE Public Data Release

Welcome to the EAGLE public data release.

Below we provide access to the EAGLE public database, which contains galaxy properties (such as masses, star formation rates, luminosities and metallicities), merger histories and images for more than 1,000,000 simulated galaxies extracted from multiple simulations of various box sizes, numerical resolutions and physical models. Additionally, we also provide access to the particle data for each of these simulations.

Access to both the database and particle data require only a single account, which you can register for below.



- Virgo's Millennium Simulation (2005) was the first to make all snapshots publicly available
- More papers published independently of the original team than by the original team
- Demonstrates ~20 year scientific value



FAIR assessment

Ainsworth 2022

- Virgo Eagle found to be 95% FAIR – 100% interoperable
- Standardized knowledge representation using formal, accessible, and broadly applicable language
- Flexible, widely supported data formats: SQL, CSV, HDF5, VOTable
- Familiar Database Structure: adopted the Millennium simulation SQL database framework
- Unique identification and interconnectivity: implemented "GalaxyID" as a unique identifier
- Supports easy data sharing and cross-referencing: contextual information provided through linked publications

EAGLE galaxy database

If you already have an account, follow this [link](#) to access the data or you can register on the following [webpage](#).

Documentation can be found on the database itself or in the galaxy catalogue [release paper](#).

The python examples from the release paper are available here:

- To connect to the database directly via Python [here](#)
- Galaxy stellar mass function example [here](#).

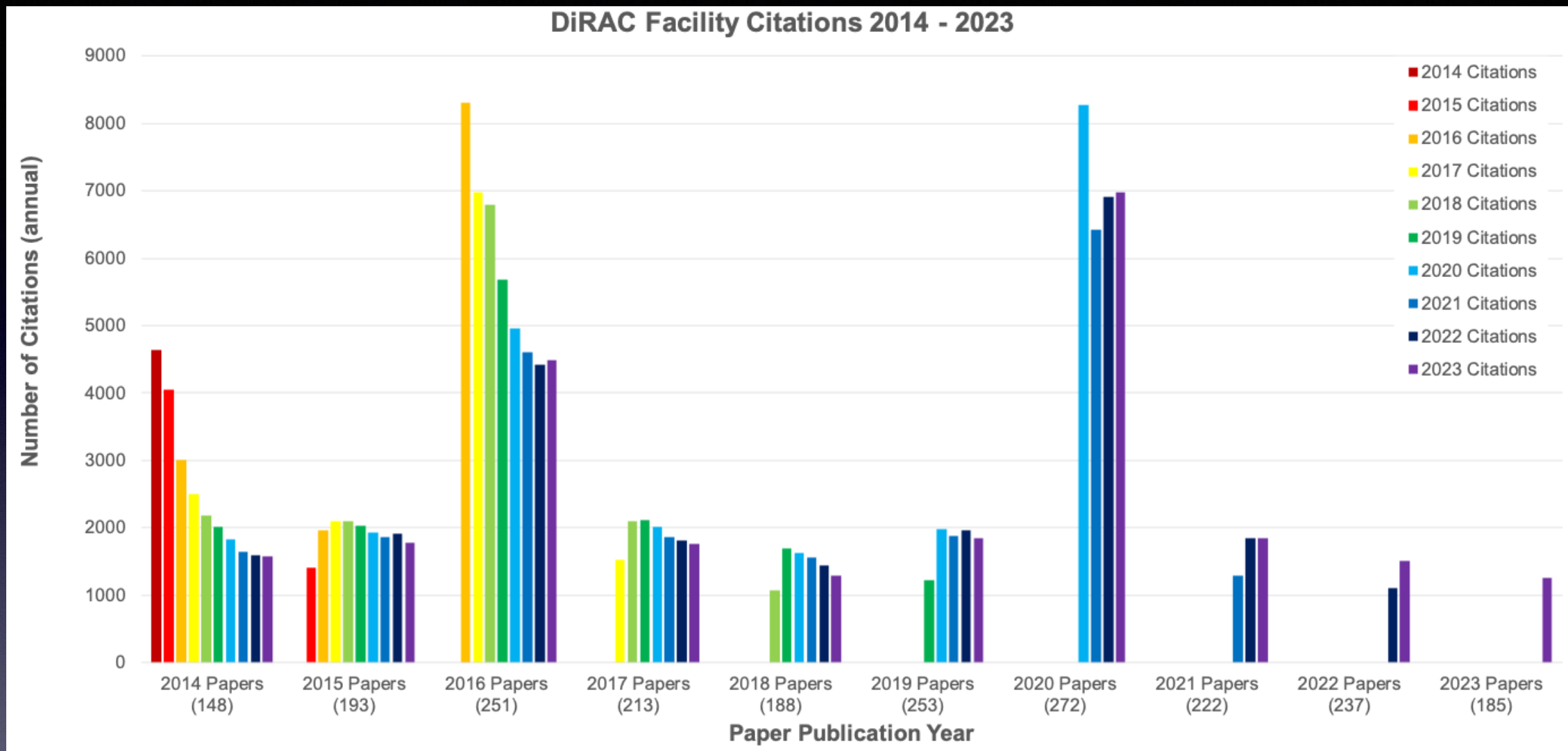
EAGLE particle data

If you already have an account, the particle data can be accessed from this [link](#).

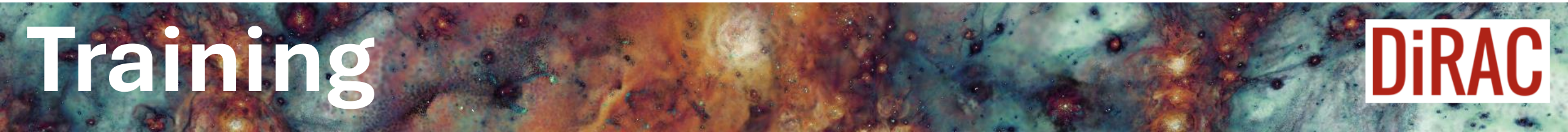
The documentation accompanying the particle data can be downloaded in the data [release paper](#).

The Python examples from the release paper are available below:

- Reading a dataset (Section 4.1) [here](#).
- Reading the snapshot header (Section 4.2) [here](#).
- Reading dark matter mass (Section 4.3) [here](#).
- Plotting the rotation curve of a galaxy (Section 4.4) [here](#).
- Plotting the temperature-density relation for a galaxy (Section 4.5) [here](#).
- The `read_eagle()` routine (Section 4.6) is available via git repository [here](#).
- The example using `read_eagle()` to create the temperature-density relation for a galaxy (Section 4.6.1) can be found [here](#).



- **NB: acknowledge DiRAC in all publications that use DiRAC resources or DiRAC-generated data**
- Tell us (early) if you're planning a press release with your University press office
 - We can help



DiRAC Training Academy

<https://dirac.ac.uk/training/>

- Comprehensive structured programme of training and support
- Academic- and industry-focused opportunities
- In-person and remote events
- Advertised via All User email and on our social media platforms
- Fosters development of skills at all levels

Foundation HPC Skills Course

- Gateway to HPC focusing on key essential skills:
 - ❖ Unix Environment
 - ❖ Command Scripts
 - ❖ Version Control
 - ❖ Software Engineering
 - ❖ Testing and Code Scaling
- All material is available self-paced on our website



Introduction to SLURM

Mastering the queuing system; Efficiently Manage Job Submission; Optimize with Module Commands; Explore a range of HPC Job Types

Introduction to MPI

First Steps in MPI; Point-to-point, Collective and Non-Blocking Communications; Porting Serial Code and Optimising MPI applications

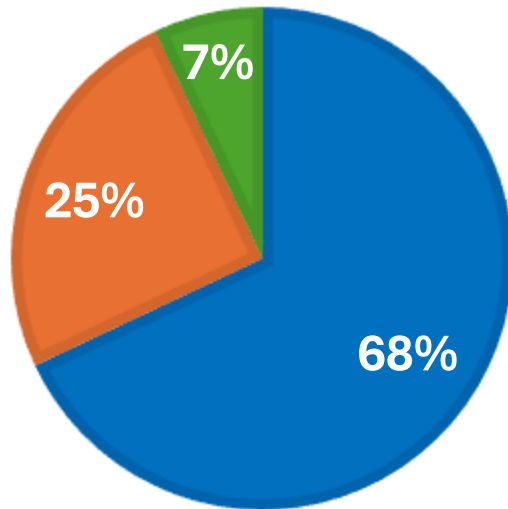
Introduction to OpenMP

Foundation of Parallelism; First Steps in OpenMP; Building Parallel Applications; Mastering Synchronization and hybrid Parallelism

DiRAC User Survey 2024

SURVEY RESPONDENT'S CODE BASE

- CPU-Centric
- CPU+GPU enabled
- GPU only



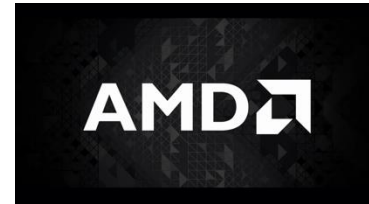
DiRAC Performance Portability Training Programme

- Currently under development
- Industry partners:



NVIDIA

Broad overview of different GPU pathways available



Advanced OpenMP and introduction to HIP



Hardware agnostic approach to porting code with oneAPI

www.dirac.ac.uk/training

Joint-Industry Training



Unleashing the Power of AMD's MI300 Series

Discover the Potential

❖ Introductory Training: March 11-12th 2025; Online

Mastering the latest GPU architecture

❖ In-Person Hackathon: April 1st – 3rd 2025; London

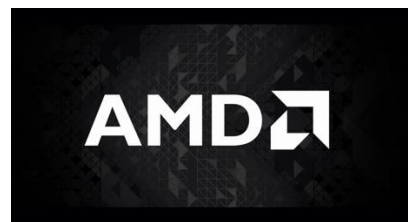
Fundamentals of Deep Learning

❖ Online Workshop: Q2 2025

Advanced GPU Programming

❖ Online Introductory Training: Q3 2025

❖ In-Person 3-day Hackathon: Q3 2025



Innovation Placement Scheme



Recent Placements



**QUANTIFYING THE EFFECTIVENESS
OF MEDICINES: LINKING HEALTH
CARE OUTCOME DATA TO
MEDICINE DATA**



Guy's and St Thomas'
NHS Foundation Trust

**IMPROVING HEALTH OUTCOMES
FOR CHILDREN AND YOUNG
PEOPLE USING DATA SCIENCE
AND AI**



**ACCELERATING AI
PIPELINES TO
ACCELERATE SCIENCE**



**Hewlett Packard
Enterprise**

**INNOVATION PLACEMENT
IN QUANTUM COMPUTING**

DiRAC Innovation Programme

DiRAC

- Broad set of activities supporting science programme and delivering benefits to external partners and/or STFC/UKRI
 - Key route to impact and benefits delivery for capital investments
 - Partners include HPC vendors, NHS, cloud providers, etc.
- Activities include silicon-level co-design, system-level co-design, deployment of test-bed systems, student/PDRA innovation placements, software development, applications of algorithms in other fields
 - Many activities support future UKRI/STFC Digital Research Infrastructure plans (e.g. DiRAC Federation Project)
- All activities support two-way knowledge transfer



DiRAC Research Software Engineering Team

computational

algorithmic

“With great  power comes great  responsibility” (Longfei Gao, KAUST)

- DiRAC resources cost money/carbon: we must use them efficiently
- HPC hardware performance requires more parallelism to be exposed
 - DiRAC science case cannot be met by hardware improvements alone
- DiRAC RSE team supports:
 - code improvement and re-factoring (e.g. migration to GPU)
 - energy efficiency
 - co-design & procurement
 - training
- RSE effort available via RAC process: collaborative activity – not just “out-sourcing”
- **Think about how RSE support could benefit you and/or your research group**

GPU Feasibility studies

- New activity proposed and led by the DiRAC RSE team with approval from DiRAC Board
 - 3-month RSE engagements to explore GPU
 - Reports will
 1. Identify blockers
 2. Determine what is required to overcome these
 3. Estimate effort required
 - Reports will be technically specific about algorithmic and/or implementation issues which need to be addressed to allow a move to GPU
 - Mix of targeted interventions by DiRAC and call for proposals
 - Will be used to quantify levels of effort required for code re-factoring across the community
 - Goal: significantly increase the fraction of code base that is GPU ready prior to any DiRAC-4 deployments
 - More information on this call will be available soon
- Question for today: if a GPU version of your code is possible in 5 years, what would make it possible in 2 years?

Applying for DiRAC resources

- Seedcorn proposals:
 - Up to 3 months: code/system testing; proposal preparation
 - Light-touch review process: decisions within ~1 week
 - Allocations up to 100k CPU core hours or 1k GPU hours
- Resource Allocation Committee (RAC) calls
 - Project categories: small (up to 12 months) and large (up to 3 years)
 - Annual calls – open in June; close in September
 - Next call: RAC18 – for allocations starting in April 2026
 - <https://dirac.ac.uk/getting-access/>
- RSE effort
 - Projects from 3 months to multiple years
 - 5 FTE of staff effort available
 - Apply via RAC calls – or ask about discretionary allocations

- DiRAC has led in sustainability since its inception
 - Mandated data centre PUE of 1.2 in 2011 when sector average was >1.6
 - COSMA carbon reporting e-mails to users
 - DiRAC-wide deployment of on-chip liquid cooling
 - Design process for DiRAC services means less physical infrastructure required
 - Clocking down of Tursa based on study carried out by Antonin Portelli
- Continued work towards minimising our carbon footprint
 - RSE-led proactive code profiling for energy optimisation
 - Power/carbon budgeting may soon be mandatory
 - Be prepared in case of grid power shortages
- Adopting challenging, but realistic, goals

Energy efficiency: Lattice Quantum Field Theory simulations on A100 & ATOS Sequana XH2000

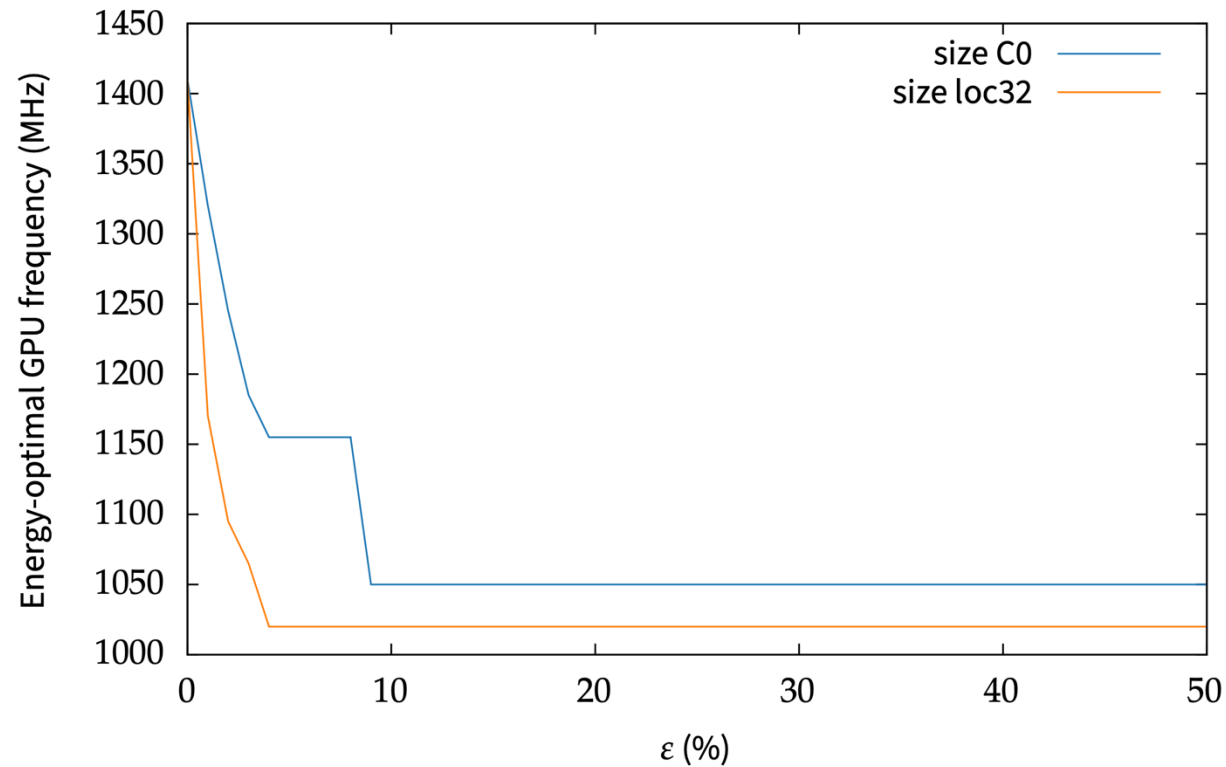
Antonin Portelli

- Study based on particle physics code: Grid
- Report commissioned by STFC DiRAC: <https://doi.org/10.5281/zenodo.7057318>
- Report data and running environment: <https://doi.org/10.5281/zenodo.7057644>

Findings:

- For the Grid workload, 16-24% energy cut possible with higher efficiency
- This is achieved with GPU frequency limit around 1 GHz
 - 10% impact on floating-point performances
 - Default frequency settings are energy-inefficient
- Lower default frequency now implemented on Tursa

Performance priority ε -constraint



ε : maximum negative impact on performances

eInfrastructure for Research and Innovation for STFC (IRIS)



IRIS is a cooperative community bringing together STFC (and beyond) computing interests

Bringing groups together
Building Infrastructure
Supporting Science
Lobbying for the future

Works closely with STFC but run by the community

Science Director: Jon Hays (QMUL)



<https://www.iris.ac.uk/>



Imperial Cloud
LondonGrid



SCARF HPC
cluster

RAL cloud

CSD3 HPC
cluster

Arcus cloud



CENTRE FOR ASTROPHYSICS



Examples of IRIS supported activities...



SKAO



CENTRE FOR ASTROPHYSICS



IRIS Activities



Build infrastructure

Invest in computing
hardware

Support digital asset creation

Build community

Support training

Making the case for investment

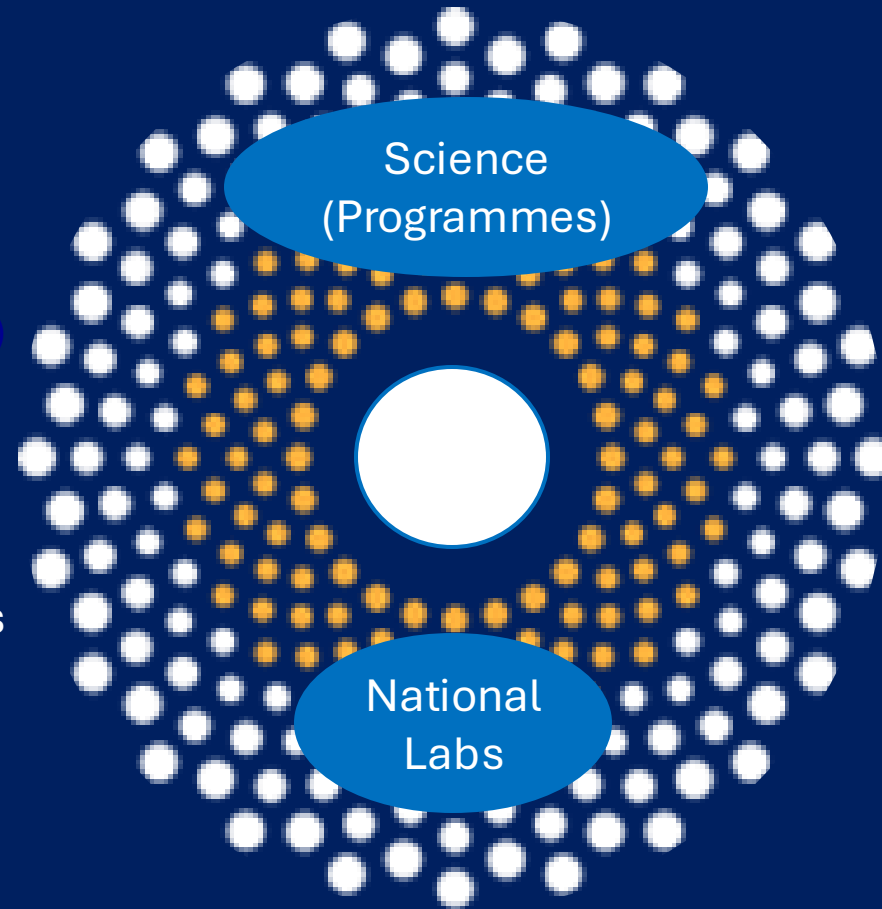


Invest in computing hardware



HTC Computing for
Experimental PP (and others)

>50k cores
> 30 high-memory nodes
> 20PB Disk
~280 GPU compute





What IRIS is not...



IRIS is not a turn-key computing solution for your favourite project

IRIS has no direct resource funding for user support – some community support as “best effort”

Physical resources for groups who can make use of it – depending on size you need your own:

- Software frameworks

- Support staff

IRIS might not have the kind of hardware you need

If you think this might be for you – despite the caveats – contact your closest Delivery Board Member

<https://www.iris.ac.uk/partners/deliveryboard/>

AI Research Resource: Isambard-AI

Phase 1

Arrived & accepted in March 2024
(Isambard 3 MDC)

1 x DLC EX2500 cabinet

21 blades (4-way Grace-Hopper)

42 nodes

168 GH superchips

12,096 Neoverse V2 Armv9 CPU cores

168 Hopper GPUs

21.5 TB CPU memory

16.1 TB high bandwidth GPU memory

37.6 TB total memory

AI high performance storage

~1 PB all-flash ClusterStor Lustre

Phase 2

New Isambard-AI ~5MW MDC Delivery of AI services (s)

12 x DLC EX4000 cabinets

660 blades (4-way Grace-Hopper)

1,320 nodes

5,280 GH superchips

380,160 Neoverse V2 Armv9 CPU cores

5,280 Hopper GPUs

675 TB CPU memory

506 TB high bandwidth GPU memory

1.18 PB total memory

AI high performance storage

~27 PB all-flash storage!

~20 PB Lustre, ~7 PB software defined VAST

AI Research Resource: DAWN

Hardware

- 256 Dell PowerEdge XE9640 nodes with DLC to CPUs and GPUs
- 2x 48 cores Intel Xeon 4th gen (Sapphire Rapids) CPU
- 4x Intel Data Center Max GPU 1550 (Ponte Vecchio) GPUs
- 4x HDR 200Gb/s per node
- **Number 41** on Top 500 in November 2023 (19.46 PFlops) with 243 nodes (65th in Nov 2024 list)

Key Design Principles

- OpenStack used to deploy DAWN nodes into CSD3 Slurm which is a Tier 2 National HPC service.
- Cloud-Native Multi-Tenancy: Supports various user environments (Jupyter, MLOps, SLURM).
- Energy Efficiency: Low environmental impact with advanced cooling technologies.

DAWN in Cambridge Data Centre



Convergence of Cloud and HPC on Isambard-AI

Application	AI and ML Applications and Frameworks					
Environment	NVIDIA Containers Standard conda / pip environments Custom conda / pip environments Install / compile your own software					
Interface	Notebooks and Dashboards			Job Scripts and Graphical Interfaces		
Platform	JupyterHub	Kubeflow	Custom Platforms	Batch Jobs	Container Runtimes	VSCode
	Kubernetes			Shell access (slurm)		
Tenancy	Multi-tenant Partitions					
Infrastructure	CSM – Cloud Native Supercomputing					

How to apply for access to AIRR

- Open to all UKRI researchers
- Projects must be AI-focused
- Call for access to Dawn and Isambard-AI (Phase 1):

<https://engagementhub.ukri.org/ukri-infrastructure/airr-eoi/>

- Isambard-AI Phase 2 will come online during summer 2025

Other UKRI computing resources

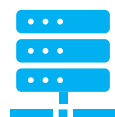
- Archer2 – 5,860 AMD Rome CPU nodes – currently only EPSRC/NERC
- Isambard-3 – 380 NVIDIA Grace-Grace CPU nodes
- Bede: 32x IBM AC922 nodes with 0.5TB and 4x32GB V100 GPU
- Apply via UKRI ***Access to HPC facilities*** process – two calls per year
- Projects up to 12 months duration
- Current call - **closing date 22nd May:**

<https://www.ukri.org/opportunity/access-to-high-performance-computing-facilities-spring-2025/>

THE EUROHPC SUPERCOMPUTING ECOSYSTEM



PETASCALE



PRE-EXASCALE

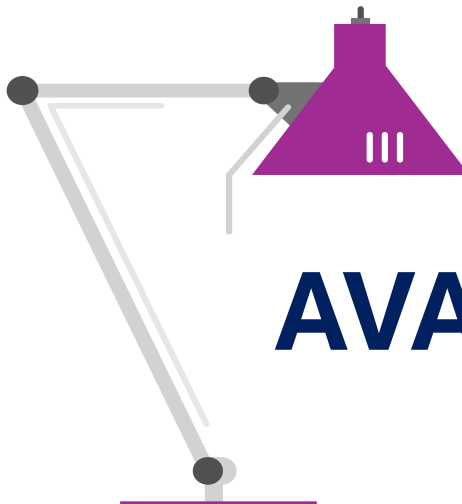


MID-RANGE



EXASCALE





AVAILABLE ACCESS MODES

PREPARATORY ACTIVITIES

BENCHMARK ACCESS

Designed for code scalability tests

Access duration: 3 months

DEVELOPMENT ACCESS

Designed for code development and optimization

Access duration: 12/6 months

PRODUCTION ACTIVITIES

EXTREME SCALE ACCESS

Designed for high-impact, high-gain research applications requesting extremely large allocations

Access duration: 12 months

REGULAR ACCESS

Designed for research applications requesting large allocations

Access duration: 12 months

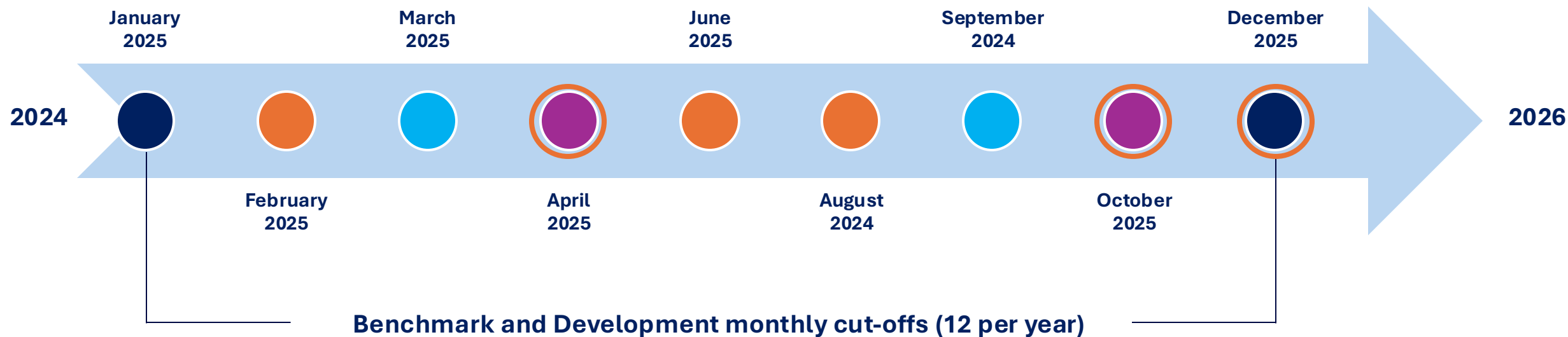
AI & DATA-INTENSIVE APPLICATIONS ACCESS

Designed for performing AI-related activities

Access duration: 12/6 months



ACCESS CALLS CUT-OFFS IN 2025



Regular Access
28 March 2025

AI & Data-Intensive Applications Access
11 April 2025

Extreme Scale Access
18 April 2025

IMPLEMENTED PEER-REVIEW PROCESS

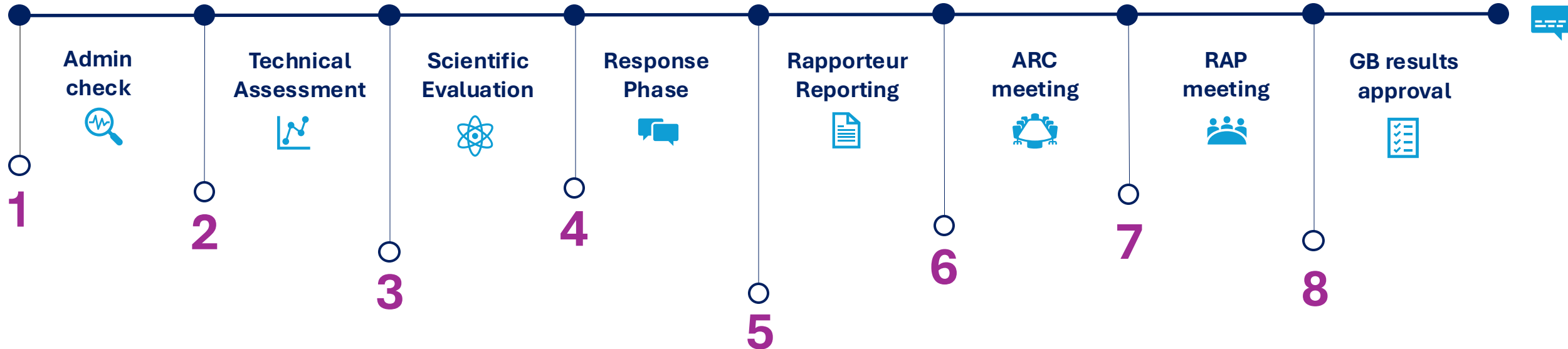


EXTREME SCALE ACCESS

REGULAR ACCESS

AI & DATA-INTENSIVE APPLICATIONS
ACCESS

BENCHMARK &
DEVELOPMENT ACCESS



EVALUATION CRITERIA

EXCELLENCE

- Proposal concept and objectives
- Expected outcomes significance
- Scientific and technical maturity
- Methodology and feasibility
- Resources request justification
- Relevance for EU research

INNOVATION & IMPACT

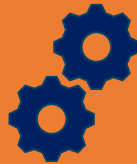
- Objectives originality and novelty
- Innovative nature of the proposed work
- Potential impact on societal, economic and/or technological dimensions

QUALITY & EFFICIENCY

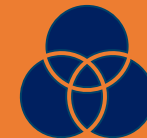
- Project plan feasibility
- Requested resources and proposed objectives alignment
- Gantt chart tasks distribution
- Competencies of the team members



WHAT?



WHY?



HOW?

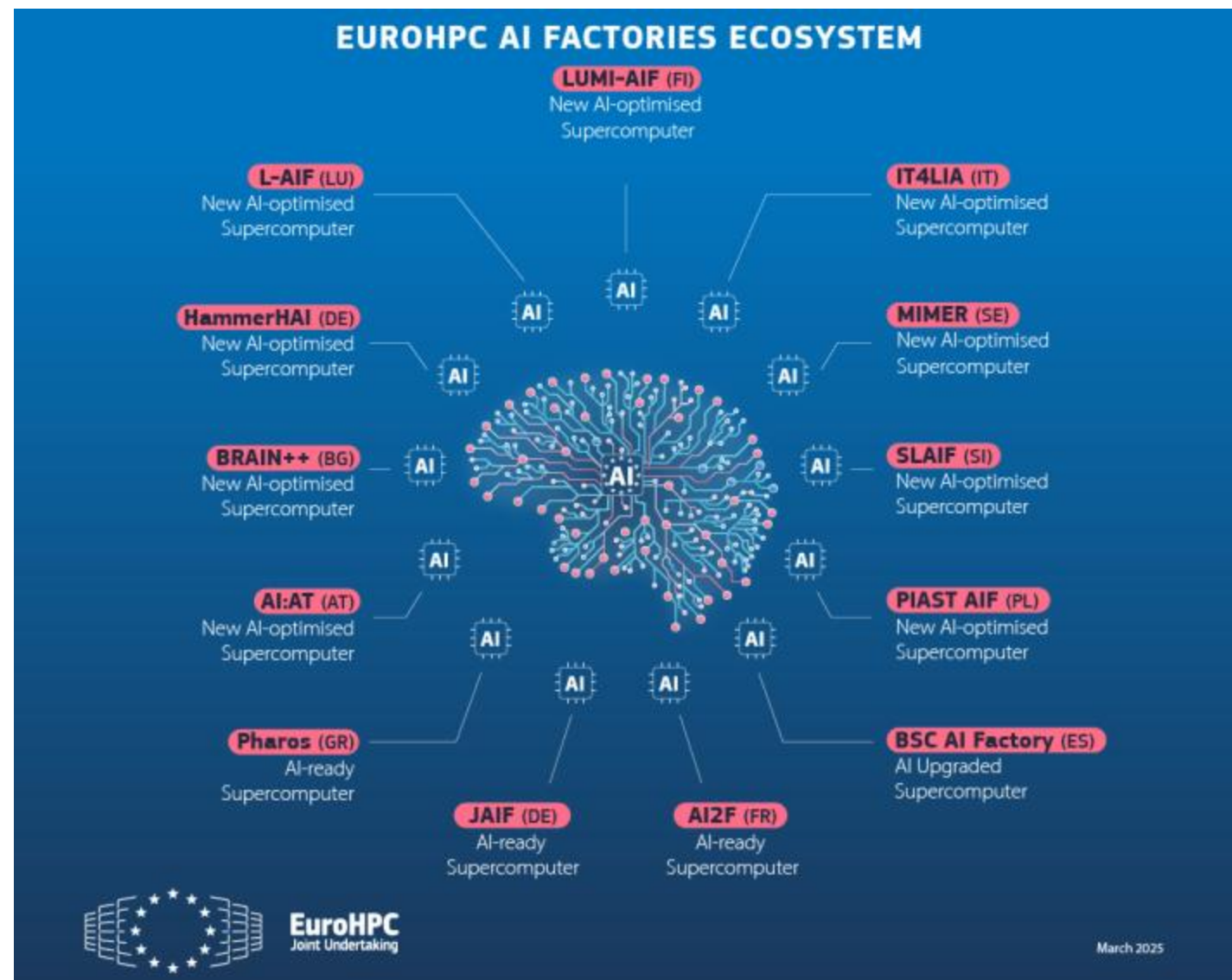
Advice from a EuroHPC ARC chair

A good proposal needs to have

- ... solid motivations
 - ... ambitious goals
 - ... performant software
 - ... requests that are justified by milestones
-
- all of this needs to be **in the application!**
 - ... needs to be **clearly stated** for external reviewers
 - ... and for a majority of panel members

THE AI FACTORIES

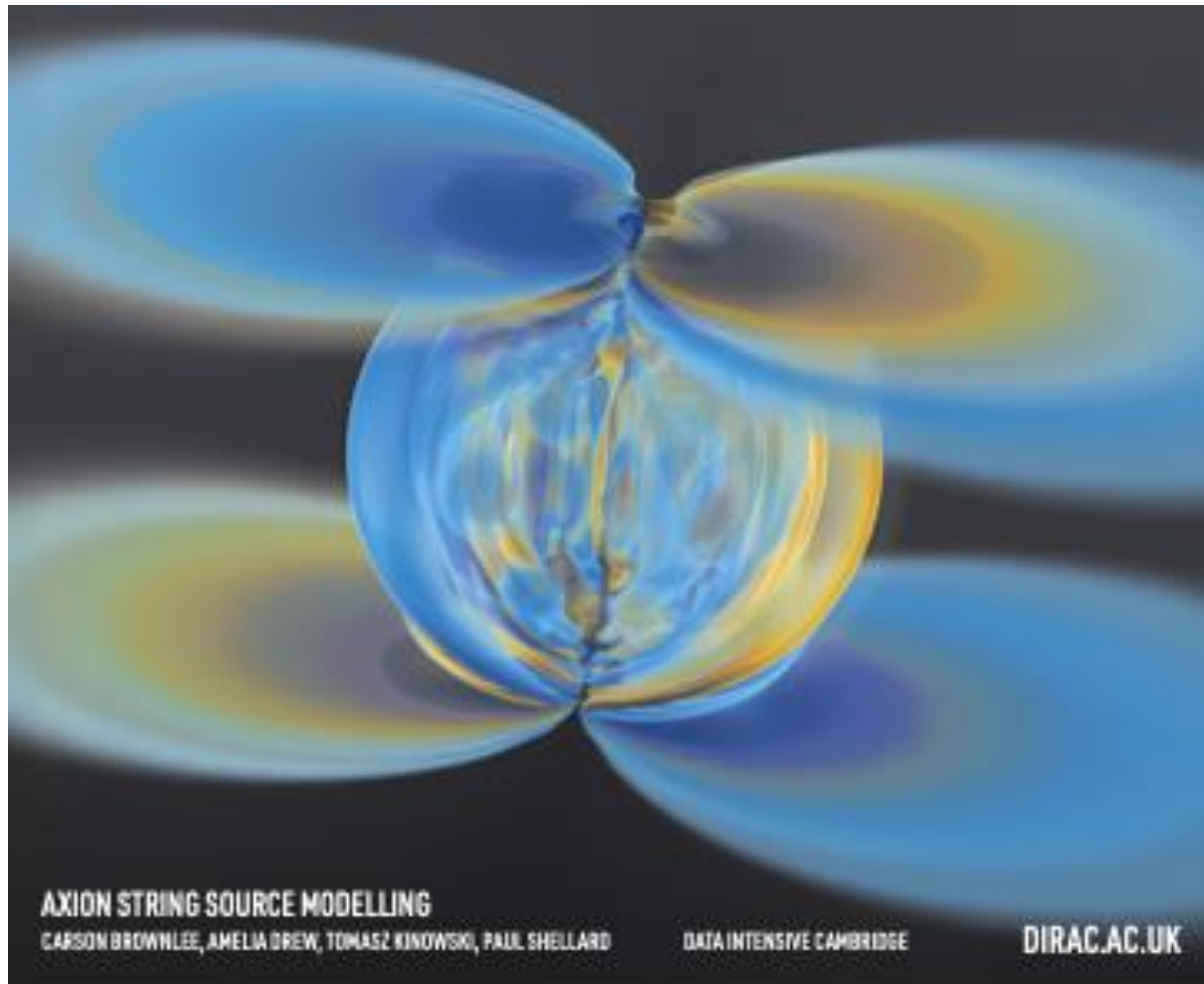
EuroHPC JU selected 13 EU sites that will host the first AI Factories – which will drive Europe's leadership in AI



DiRAC Image Competition 2024 - the winners

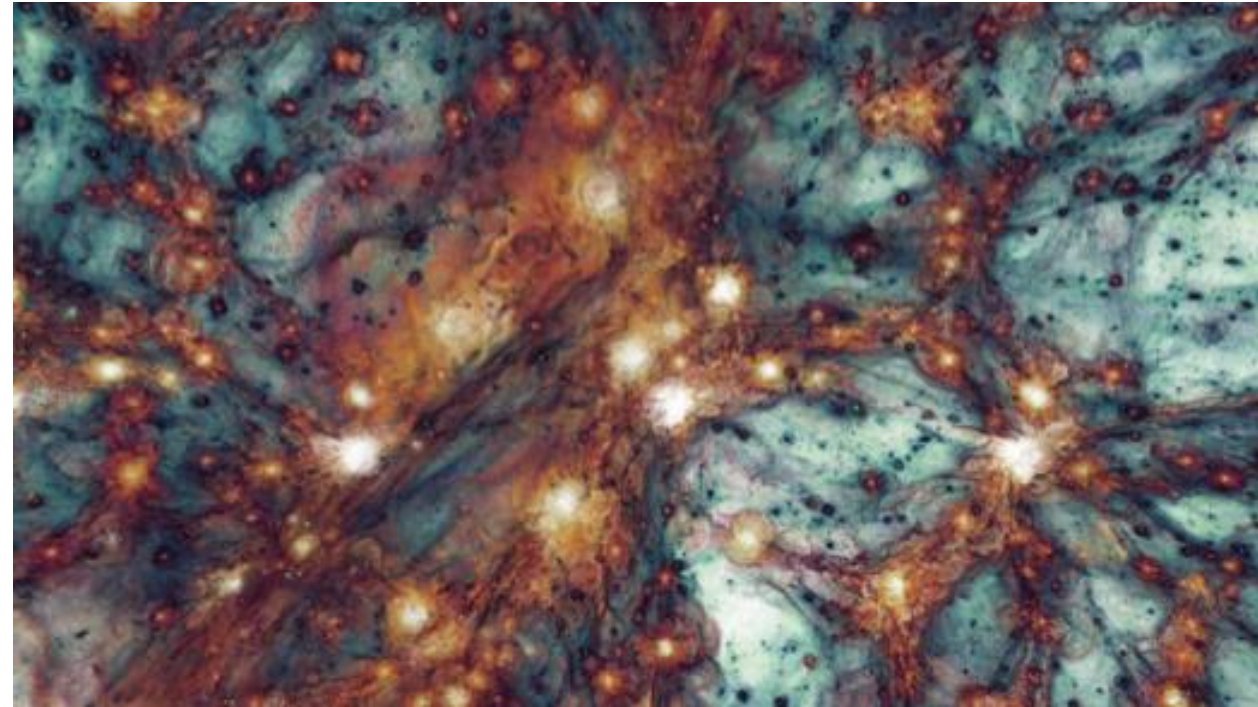
Axion String Source Modelling

Carson Brownlee, Amelia Drew,
Tomasz Kinowski, Paul Shellard



A “Shocking” View Of A Visual Universe

The FLAMINGO and COLIBRE teams



Summary

- **DiRAC resources are intended to support your science**
 - Having problems? Submit a support ticket
 - Remember to acknowledge DiRAC in your papers
 - Get involved in the DiRAC-4 co-design process
- **Take advantage of all training opportunities**
- **Is your code as efficient as it could be? Is it portable to GPUs?**
 - Training and RSE support available to help you improve your code and reduce the carbon footprint of your research
- **DiRAC can help you develop industry contacts**
- **Give us your feedback – positive or negative!**
 - DiRAC User survey

More information....

- **DiRAC:**
 - www.dirac.ac.uk
 -  DiRAC - HPC
 - miw6@leicester.ac.uk
 - **DiRAC Day 2025:** Stoke-on-Trent, 11th December
- **IRIS:** <https://www.iris.ac.uk/partners/deliveryboard/>
- **UKRI Tier2:** <https://www.ukri.org/opportunity/access-to-high-performance-computing-facilities-spring-2025/>
- **AIRR:** <https://engagementhub.ukri.org/ukri-infrastructure/airr-eoi/>
- **EuroHPC:** https://eurohpc-ju.europa.eu/supercomputers/supercomputers-access-calls_en